

Inputs: $\mathbf{X}$, $\mathbf{o}$, $\mathbf{w}$, ϵ_R , ϵ_m

1. Measure

$R_j, \|\mathbf{x}_j\|_2$ per column as published

$R_j \leq \epsilon_R$?

yes

Move j to $\mathcal{Z}$
(takes no further part)

no

$\kappa_j = R_j / \|\mathbf{x}_j\|_2$ for $j \in \mathcal{J}$

2. Reweight

$\tilde{w}_j = w_j \kappa_j / \sum_{j'} w_{j'} \kappa_{j'}$

3. Summarize

$\nu = \max_j \kappa_j / \min_j \kappa_j$,
 $\delta_w = \frac{1}{2} \sum_j |\tilde{w}_j - w_j|$

4. Screen pairs

$\sigma_1 = \sum_j a_j$, $\sigma_2 = \sum_j \kappa_j a_j$

signs of σ_1, σ_2 differ,
 $|\sigma_1|, |\sigma_2| > \epsilon_m$?

yes

Add pair (i, i') to
warning set $\mathcal{P}$

no

Audit report

$\kappa, \tilde{\mathbf{w}}, \nu, \delta_w, \mathcal{P}, \mathcal{Z}$

5. Compensate

$w_j^* = (w_j / \kappa_j) / \sum_{j'} w_{j'} / \kappa_{j'}$

Ranking tool (unchanged):
WSM, TOPSIS, ...
run with $\mathbf{w}$ or $\mathbf{w}^*$